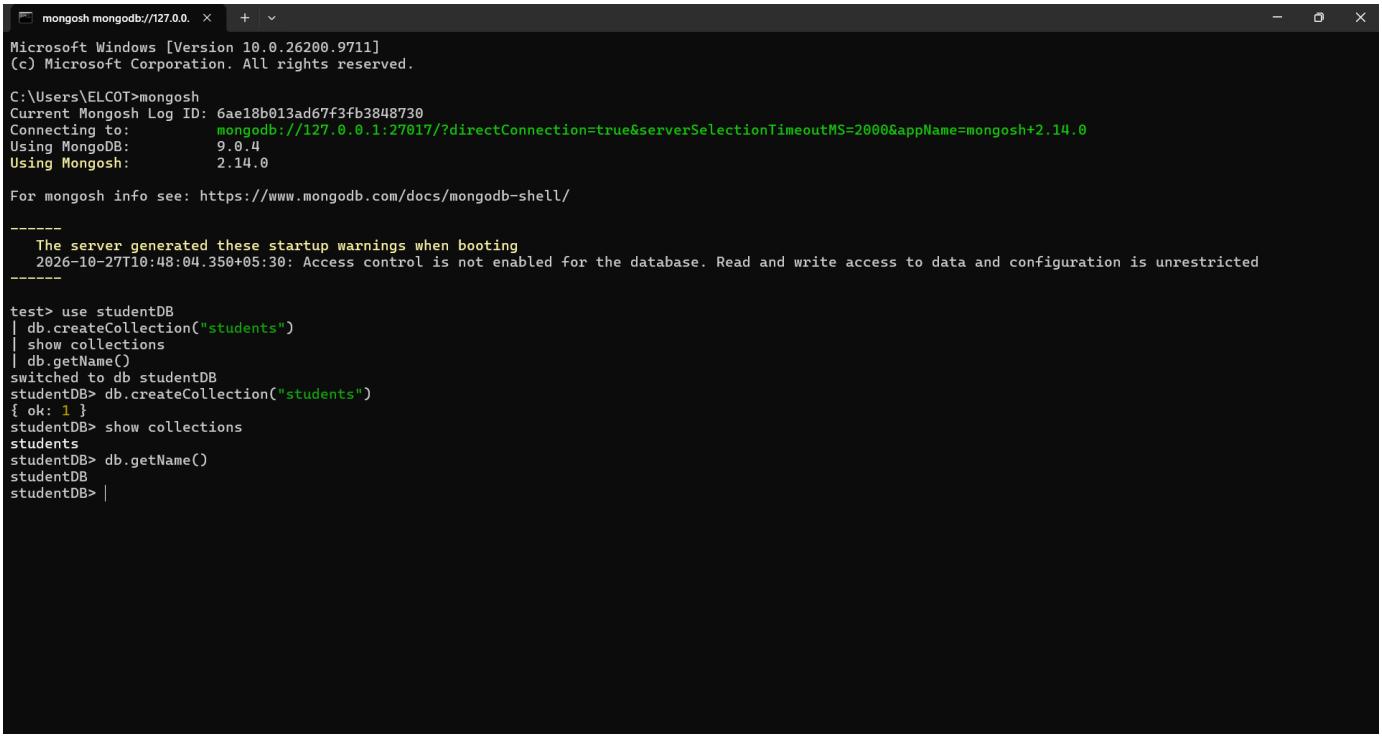
MongoDB CRUD Operations

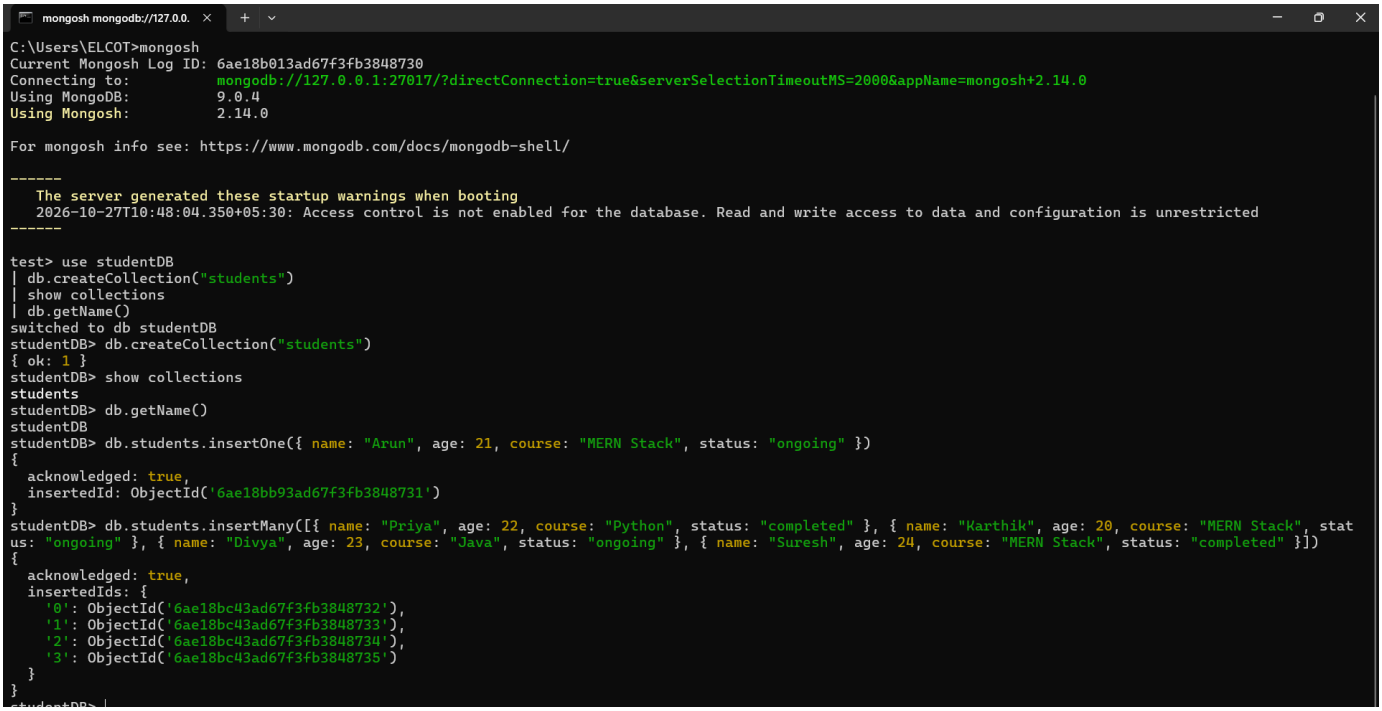
Week 12 - Terminal Practice Screenshots

Databases: studentDB, ecommerceDB

Step 1: Database and Collection Setup (use studentDB, createCollection, show collections)



Step 2: Insert Operations (insertOne and insertMany)



Step 3a: Read - fetch all documents (find)

```
  }
}
studentDB> db.students.find()
[
  {
    _id: ObjectId('6ae18bb93ad67f3fb3848731'),
    name: 'Arun',
    age: 21,
    course: 'MERN Stack',
    status: 'ongoing'
  },
  {
    _id: ObjectId('6ae18bc43ad67f3fb3848732'),
    name: 'Priya',
    age: 22,
    course: 'Python',
    status: 'completed'
  },
  {
    _id: ObjectId('6ae18bc43ad67f3fb3848733'),
    name: 'Karthik',
    age: 20,
    course: 'MERN Stack',
    status: 'ongoing'
  },
  {
    _id: ObjectId('6ae18bc43ad67f3fb3848734'),
    name: 'Divya',
    age: 23,
    course: 'Java',
    status: 'ongoing'
  },
  {
    _id: ObjectId('6ae18bc43ad67f3fb3848735'),
    name: 'Suresh',
    age: 24,
    course: 'MERN Stack',
    status: 'completed'
  }
]
studentDB>
```

Step 3b: Read - filter by course MERN Stack

```
]
studentDB> db.students.find({ course: "MERN Stack" })
[
  {
    _id: ObjectId('6ae18bb93ad67f3fb3848731'),
    name: 'Arun',
    age: 21,
    course: 'MERN Stack',
    status: 'ongoing'
  },
  {
    _id: ObjectId('6ae18bc43ad67f3fb3848733'),
    name: 'Karthik',
    age: 20,
    course: 'MERN Stack',
    status: 'ongoing'
  },
  {
    _id: ObjectId('6ae18bc43ad67f3fb3848735'),
    name: 'Suresh',
    age: 24,
    course: 'MERN Stack',
    status: 'completed'
  }
]
```

Step 3c: Read - filter by status completed

```
studentDB> db.students.find({ status: "completed" })
[
  {
    _id: ObjectId('6ae18bc43ad67f3fb3848732'),
    name: 'Priya',
    age: 22,
    course: 'Python',
    status: 'completed'
  },
  {
    _id: ObjectId('6ae18bc43ad67f3fb3848735'),
    name: 'Suresh',
    age: 24,
    course: 'MERN Stack',
    status: 'completed'
  }
]
studentDB>
```

Step 4: Update Operations (updateOne and updateMany)

```
studentDB> db.students.updateOne({ name: "Arun" }, { $set: { status: "completed" } })
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
studentDB> db.students.updateMany({ course: "MERN Stack" }, { $set: { batch: "2026" } })
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 3,
  modifiedCount: 3,
  upsertedCount: 0
}
studentDB> db.students.find({ course: "MERN Stack" })
[
  {
    _id: ObjectId('6ae18bb93ad67f3fb3848731'),
    name: 'Arun',
    age: 21,
    course: 'MERN Stack',
    status: 'completed',
    batch: '2026'
  },
  {
    _id: ObjectId('6ae18bc43ad67f3fb3848733'),
    name: 'Karthik',
    age: 20,
    course: 'MERN Stack',
    status: 'ongoing',
    batch: '2026'
  },
  {
    _id: ObjectId('6ae18bc43ad67f3fb3848735'),
    name: 'Suresh',
    age: 24,
    course: 'MERN Stack',
    status: 'completed',
    batch: '2026'
```

Step 5a: Delete one document (deleteOne)

```
studentDB> db.students.deleteOne({ name: "Divya" })
{ acknowledged: true, deletedCount: 1 }
studentDB> db.students.find()
[
  {
    _id: ObjectId('6ae18bb93ad67f3fb3848731'),
    name: 'Arun',
    age: 21,
    course: 'MERN Stack',
    status: 'completed',
    batch: '2026'
  },
  {
    _id: ObjectId('6ae18bc43ad67f3fb3848732'),
    name: 'Priya',
    age: 22,
    course: 'Python',
    status: 'completed'
  },
  {
    _id: ObjectId('6ae18bc43ad67f3fb3848733'),
    name: 'Karthik',
    age: 20,
    course: 'MERN Stack',
    status: 'ongoing',
    batch: '2026'
  },
  {
    _id: ObjectId('6ae18bc43ad67f3fb3848735'),
    name: 'Suresh',
    age: 24,
    course: 'MERN Stack',
    status: 'completed',
    batch: '2026'
  }
]
studentDB>
```

Step 6a and 6b: Query operators $gt and $lt

```
studentDB> db.students.find({ age: { $gt: 21 } })
[
  {
    _id: ObjectId('6ae18bc43ad67f3fb3848732'),
    name: 'Priya',
    age: 22,
    course: 'Python',
    status: 'completed'
  },
  {
    _id: ObjectId('6ae18bc43ad67f3fb3848735'),
    name: 'Suresh',
    age: 24,
    course: 'MERN Stack',
    status: 'completed',
    batch: '2026'
  }
]
studentDB> db.students.find({ age: { $lt: 22 } })
[
  {
    _id: ObjectId('6ae18bb93ad67f3fb3848731'),
    name: 'Arun',
    age: 21,
    course: 'MERN Stack',
    status: 'completed',
    batch: '2026'
  },
  {
    _id: ObjectId('6ae18bc43ad67f3fb3848733'),
    name: 'Karthik',
    age: 20,
    course: 'MERN Stack',
    status: 'ongoing',
    batch: '2026'
  }
]
```

Step 6c: Query operator $in

```
studentDB> db.students.find({ course: { $in: ["Python", "MERN Stack"
[
  {
    _id: ObjectId('6ae18bb93ad67f3fb3848731'),
    name: 'Arun',
    age: 21,
    course: 'MERN Stack',
    status: 'completed',
    batch: '2026'
  },
  {
    _id: ObjectId('6ae18bc43ad67f3fb3848732'),
    name: 'Priya',
    age: 22,
    course: 'Python',
    status: 'completed'
  },
  {
    _id: ObjectId('6ae18bc43ad67f3fb3848733'),
    name: 'Karthik',
    age: 20,
    course: 'MERN Stack',
    status: 'ongoing',
    batch: '2026'
  },
  {
    _id: ObjectId('6ae18bc43ad67f3fb3848735'),
    name: 'Suresh',
    age: 24,
    course: 'MERN Stack',
    status: 'completed',
    batch: '2026'
  }
]
```

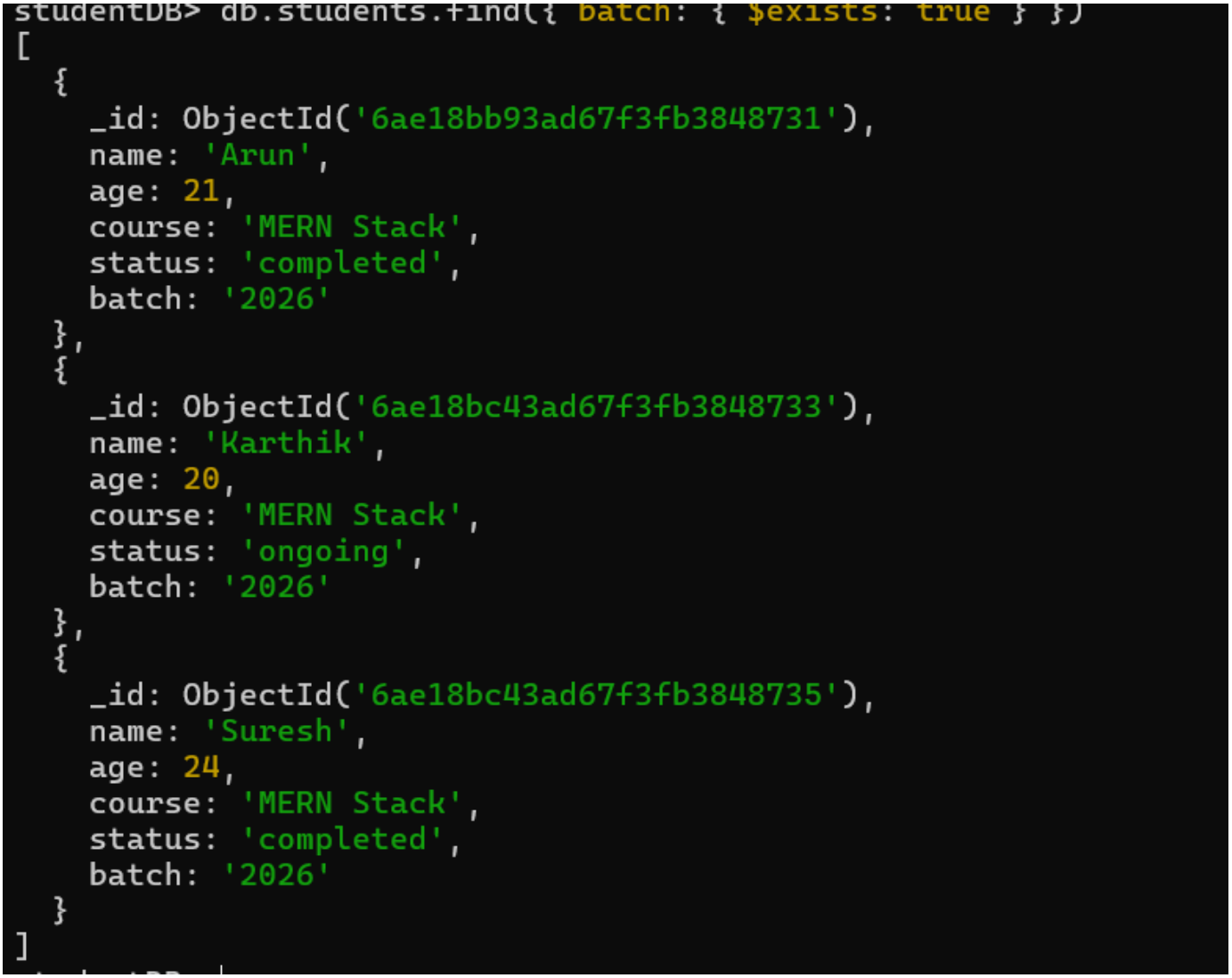
Step 6d: Query operator $and

```
studentDB> db.students.find({ $and: [{ course: "MERN Stack" }, { status: "completed" }] })
[
  {
    _id: ObjectId('6ae18bb93ad67f3fb3848731'),
    name: 'Arun',
    age: 21,
    course: 'MERN Stack',
    status: 'completed',
    batch: '2026'
  },
  {
    _id: ObjectId('6ae18bc43ad67f3fb3848735'),
    name: 'Suresh',
    age: 24,
    course: 'MERN Stack',
    status: 'completed',
    batch: '2026'
  }
]
studentDB>
```

Step 6e: Query operator $or

```
studentDB> db.students.find({ $or: [{ course: "Python" }, { status: "ongoing" }]
[
  {
    _id: ObjectId('6ae18bc43ad67f3fb3848732'),
    name: 'Priya',
    age: 22,
    course: 'Python',
    status: 'completed'
  },
  {
    _id: ObjectId('6ae18bc43ad67f3fb3848733'),
    name: 'Karthik',
    age: 20,
    course: 'MERN Stack',
    status: 'ongoing',
    batch: '2026'
  }
]
studentDB> db.students.find({ batch: { $exists: true } })
[
  {
    _id: ObjectId('6ae18bb93ad67f3fb3848731'),
    name: 'Arun',
    age: 21,
    course: 'MERN Stack',
    status: 'completed',
    batch: '2026'
  },
  {
    _id: ObjectId('6ae18bc43ad67f3fb3848733'),
    name: 'Karthik',
    age: 20,
    course: 'MERN Stack',
    status: 'ongoing',
    batch: '2026'
  },
  {
    _id: ObjectId('6ae18bc43ad67f3fb3848735'),
    name: 'Suresh',
    age: 24,
```

Step 6f: Query operator $exists



Step 5b: Delete all documents (deleteMany) - practice

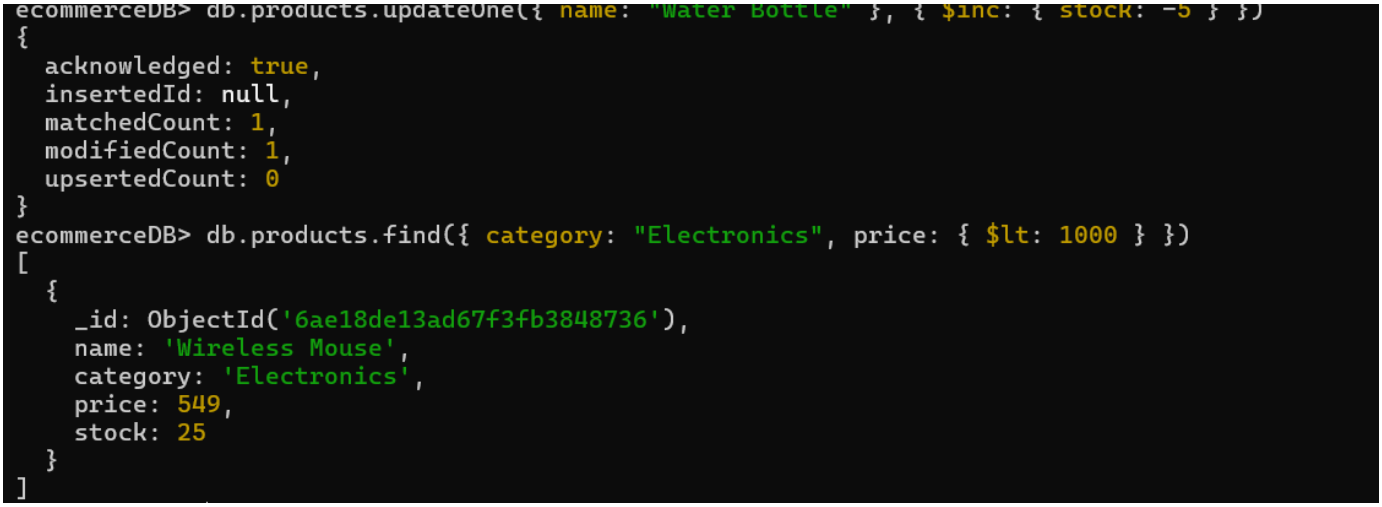
```
studentDB> db.students.find({ batch: { $exists: true } })
[
  {
    _id: ObjectId('6ae18bb93ad67f3fb3848731'),
    name: 'Arun',
    age: 21,
    course: 'MERN Stack',
    status: 'completed',
    batch: '2026'
  },
  {
    _id: ObjectId('6ae18bc43ad67f3fb3848733'),
    name: 'Karthik',
    age: 20,
    course: 'MERN Stack',
    status: 'ongoing',
    batch: '2026'
  },
  {
    _id: ObjectId('6ae18bc43ad67f3fb3848735'),
    name: 'Suresh',
    age: 24,
    course: 'MERN Stack',
    status: 'completed',
    batch: '2026'
  }
]
studentDB> db.students.deleteMany({})
{ acknowledged: true, deletedCount: 4 }
studentDB> db.students.find()

studentDB>
```

Step 7a and 7b: E-Commerce use case - insert products, update price

```
studentDB> use ecommerceDB
switched to db ecommerceDB
ecommerceDB> db.products.insertMany([{ name: "Wireless Mouse", category: "Electronics", price: 599, stock: 25 }, { name: "Cotton T-Shirt", category: "Clothi
ng", price: 349, stock: 40 }, { name: "Water Bottle", category: "Home", price: 199, stock: 60 }, { name: "Bluetooth Speaker", category: "Electronics", price
: 1499, stock: 10 }])
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('6ae18de13ad67f3fb3848736'),
    '1': ObjectId('6ae18de13ad67f3fb3848737'),
    '2': ObjectId('6ae18de13ad67f3fb3848738'),
    '3': ObjectId('6ae18de13ad67f3fb3848739')
  }
}
ecommerceDB> db.products.updateOne({ name: "Wireless Mouse" }, { $set: { price: 549 } })
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
```

Step 7b and 7c: E-Commerce use case - update stock, search Electronics under 1000



```
ecommerceDB> db.products.updateOne({ name: "Water Bottle" }, { $inc: { stock: -5 } })
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
ecommerceDB> db.products.find({ category: "Electronics", price: { $lt: 1000 } })
[
  {
    _id: ObjectId('6ae18de13ad67f3fb3848736'),
    name: 'Wireless Mouse',
    category: 'Electronics',
    price: 549,
    stock: 25
  }
]
```

Step 7d: E-Commerce use case - view all products

```
ecommerceDB> db.products.find()
[
  {
    _id: ObjectId('6ae18de13ad67f3fb3848736'),
    name: 'Wireless Mouse',
    category: 'Electronics',
    price: 549,
    stock: 25
  },
  {
    _id: ObjectId('6ae18de13ad67f3fb3848737'),
    name: 'Cotton T-Shirt',
    category: 'Clothing',
    price: 349,
    stock: 40
  },
  {
    _id: ObjectId('6ae18de13ad67f3fb3848738'),
    name: 'Water Bottle',
    category: 'Home',
    price: 199,
    stock: 55
  },
  {
    _id: ObjectId('6ae18de13ad67f3fb3848739'),
    name: 'Bluetooth Speaker',
    category: 'Electronics',
    price: 1499,
    stock: 10
  }
]
```